

# Luoting (Lottie) Zhuang

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## SUMMARY

Ph.D. candidate in Medical Informatics at UCLA with 5+ years of experience developing machine learning methods for oncology, precision medicine, and translational research. Experience spans multimodal foundation models, medical imaging, clinical trial data, longitudinal modeling, biomarker discovery, and clinical outcome prediction. Built end-to-end ML workflows from data curation and preprocessing through model development, benchmarking, validation, and interpretation. Industry experience spans pharma and AI-enabled biotech, including clinical trial analytics and drug target assessment. Seeking machine learning and data science roles in translational drug research and development. Expected graduation: June 2027. Authorized to work in the U.S. without future sponsorship.

## SKILLS

- Programming & Tools:** Python (PyTorch, TensorFlow, scikit-learn), R, SQL, Git, Docker, Weights & Biases
- ML & AI:** vision-language and multimodal models, foundation models, self-supervised and contrastive learning, representation learning, VAEs, optimal transport, in-context learning, model interpretability
- Statistics & Data Science:** clinical outcome analysis, survival analysis, patient stratification, Cox regression, model calibration, bootstrapping, missing data imputation, data harmonization, hypothesis testing
- Translational Domains:** medical imaging, clinical trials, digital pathology, genomics, clinical NLP, biomarker discovery, precision oncology, drug discovery

## RELEVANT EXPERIENCE IN CLINICAL ML & DRUG DEVELOPMENT

|                             |                      |
|-----------------------------|----------------------|
| <b>Bristol Myers Squibb</b> | June 2026 – Aug 2026 |
| <b>Summer Intern</b>        | East Cambridge, MA   |

- Engineered a multimodal variational autoencoder pipeline for cross-study translation across unpaired clinical trial datasets with limited feature overlap, improving the translation performance Pearson correlation by 30% and reducing Wasserstein distance by 16%.
- Aligned features from three clinical trials in a shared latent space and identified candidate biomarkers with a significant association to clinical outcome.
- Established optimal transport baselines to establish benchmarks for cross-domain feature translation across heterogeneous clinical trial datasets.
- Presented modeling approaches, experimental results, and biological findings to cross-functional scientific teams.

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|------------------------------------|---------------------|
| <b>Hsu Lab, UCLA Health</b>        | June 2023 – present |
| <b>Graduate Student Researcher</b> | Los Angeles, CA     |

### Multimodal Foundation Models and Clinical Prediction

- Developed a CLIP-based vision-language model to learn clinically informed imaging biomarkers, outperforming SOTA by 5% AUROC and 14% AUPRC in lung cancer prediction.

### Longitudinal and Disease Progression Modeling

- Extended pseudotime inference into a novel imaging domain to reconstruct lung nodule trajectories from static, cross-sectional CT scans.
- Constructed a longitudinal dataset of pulmonary nodules by implementing detection using MONAI and image registration, enabling robust temporal modeling and progression analysis.

### Clinical Imaging Pipeline Development

- Designed a deep learning segmentation model with a masking strategy to generate complete lung masks that capture juxtapleural nodules, reducing pixel-level dependencies and improving nodule inclusion sensitivity by 11%.
- Contributed to developing biologically informed CT image registration capturing global and local anatomical features, achieving a 21% reduction in TRE for nodule tracking across longitudinal CT.

### Model Robustness and Representation Evaluation

- Utilized in-context learning in generalist segmentation models to examine the semantic consistency and robustness of various imaging features, including radiomics, foundation, and CLIP-derived representations.
- Assessed the robustness of deep imaging features against varying acquisition and reconstruction parameters, such as kernels, slice thickness, and dose levels.

## AI for Pathology Image Analysis Lab, Harvard Medical School

Jan 2021 – Aug 2022

### Research Assistant

Boston, MA

- Developed a weakly-supervised, interpretable pipeline using multiple instance learning for survival prediction in glioma and lung cancer patients from MRI and CT data.
- Proposed and evaluated multimodal fusion frameworks integrating histology, radiology, and genomics for glioma and lung cancer prognosis, identifying optimal fusion strategies for prognostic modeling.

### BenevolentAI

June 2021 – Aug 2021

### AI Scientist Summer Intern

Remote

- Built ML models to predict target safety, biological relevance, novelty, ligandability, and assayability, improving AUC by 8-20% across attributes.
- Improved the interpretability of target prediction by providing category-level and SHAP feature-level explanations.
- Collaborated with drug discovery scientists to assess model outputs for biological plausibility and decision support.

## EDUCATION

### University of California, Los Angeles

Sept 2022 – June 2027 (expected)

### Ph.D. in Medical Informatics

Los Angeles, CA

- **Dissertation:** Improved Methods for Modeling the Evolution of Pulmonary Nodules to Predict Lung Cancer (Advisor: Dr. William Hsu)

### Harvard University

Sept 2020 – Dec 2021

### M.S. in Biomedical Informatics

Boston, MA

- **Thesis:** Deep Learning-based Multimodal Integration of Genomics, Histology, and Radiology for Improved Outcome Prediction in Glioma Patients (Advisor: Dr. Faisal Mahmood)

### University of California, Los Angeles

Sept 2016 – Mar 2020

### B.S. in Applied Mathematics (Statistics Minor)

Los Angeles, CA

## SELECTED PUBLICATIONS

### Multimodal and Longitudinal Modeling

- **Zhuang L**, et al. (2026) From Cross-Sectional CT to Dynamic Insights: Pseudotime-Based Modeling of Lung Nodule Progression. Medical Imaging with Deep Learning.
- **Zhuang L**, et al. (2025) Vision-Language Model-Based Semantic-Guided Imaging Biomarker for Lung Nodule Malignancy Prediction. Journal of Biomedical Informatics.
- **Zhuang L\***, Park SH\*, et al. (2025) Advancing Precision Oncology Through Modeling of Longitudinal and Multimodal Data. IEEE Reviews in Biomedical Engineering.
- **Zhuang L**, Lipkova J, Chen R, Mahmood F. (2022) Deep Learning-Based Integration of Histology, Radiology, and Genomics for Improved Survival Prediction in Glioma Patients. Medical Imaging: Digital and Computational Pathology, SPIE.
- Lipkova J, Chen RJ, Chen B, Lu MY, Barbieri M, Shao D, Vaidya AJ, Chen C, **Zhuang L**, Williamson DF, Shaban M. (2022) Artificial Intelligence for Multimodal Data Integration in Oncology. Cancer Cell.

### Generative AI & Clinical NLP

- Zhang T, Ding R, **Zhuang L**, et al. (2026) Semi-Supervised Domain Adaptation with Latent Diffusion for Pathology Image Classification. IEEE Journal of Biomedical and Health Informatics.
- Xue Y, Zhu Y, **Zhuang L**, et al. (2025) SmokeBERT: A BERT-based Model for Quantitative Smoking History Extraction from Clinical Narratives to Improve Lung Cancer Screening. JCO Clinical Cancer Informatics.

### Medical Image Analysis

- **Zhuang L**, et al. (2025) Enhancing Lung Segmentation Algorithms to Ensure Inclusion of Juxtapleural Nodules. IEEE International Symposium on Biomedical Imaging (ISBI).
- **Zhuang L**, et al. (2025) Comparing the Characteristics and Robustness of Imaging Features via Prompt Selection in Generalist Segmentation Models. Medical Imaging: Imaging Informatics, SPIE.
- Zhu Y, Chien A, Kulkarni K, **Zhuang L**, et al. (2025) GLIDE-Reg: Global Local Image DEformable Registration. Under Review.

## PRESENTATIONS

- Delivered 10+ oral and poster presentations at major conferences and workshops (AMIA, EDRN, MIDL, IEEE BHI, IEEE EMBC, IEEE ISBI, RSNA, SPIE Medical Imaging, The Early Detection of Cancer Conference)